

Intro to AI for Medicine

Lec 5: Data

Last Time

- Learned about loss functions, finishing neural net structure
- Defined regression, classification (including multiclass)
- Associated final activation with loss function and problem type
- Discussed some metrics for measuring how good our networks are

Python Gang

Syllabus

20 lessons • 9 projects • 9 quizzes

Free Course

Learn Python 2

Learn the basics of the world's fastest growing and most popular programming language used by software engineers, analysts, data scientists, and machine learning engineers alike.

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1

Python Syntax

In this course, learn about the syntax of the Python programming language!



2

Strings and Console Output

This course will introduce you to strings and console output in Python.



3

Conditionals and Control Flow

Learn how to use conditionals and control flow to create programs that generate different outcomes.



Agenda

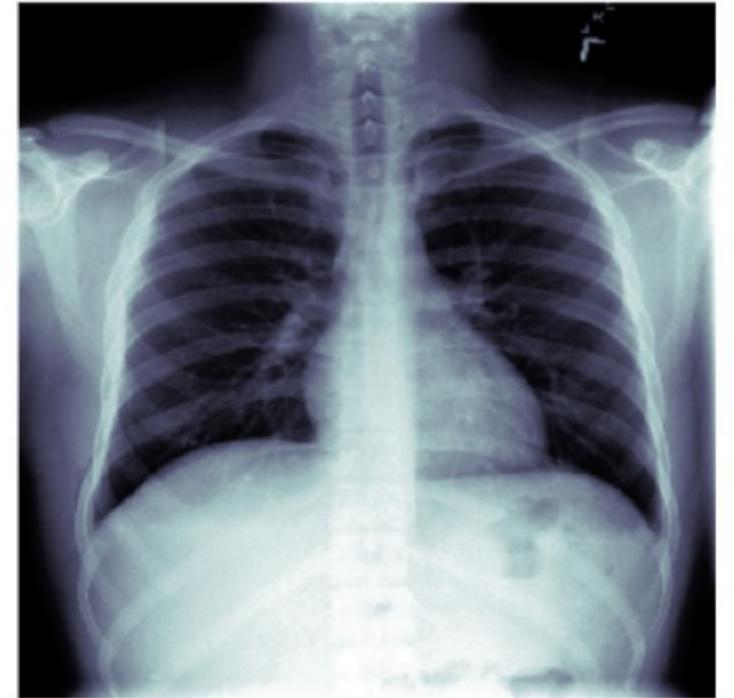
- Data representations
- Bias in data (briefly)
- Data labelling
- Data augmentation
- Medical databases out there

So Far

- We've learned the structure of the whole neural net start -> finish
- We've learned different tricks on how to train so we can get the most out of the neural net

Representing Data

- How would we represent the following in matrix format?



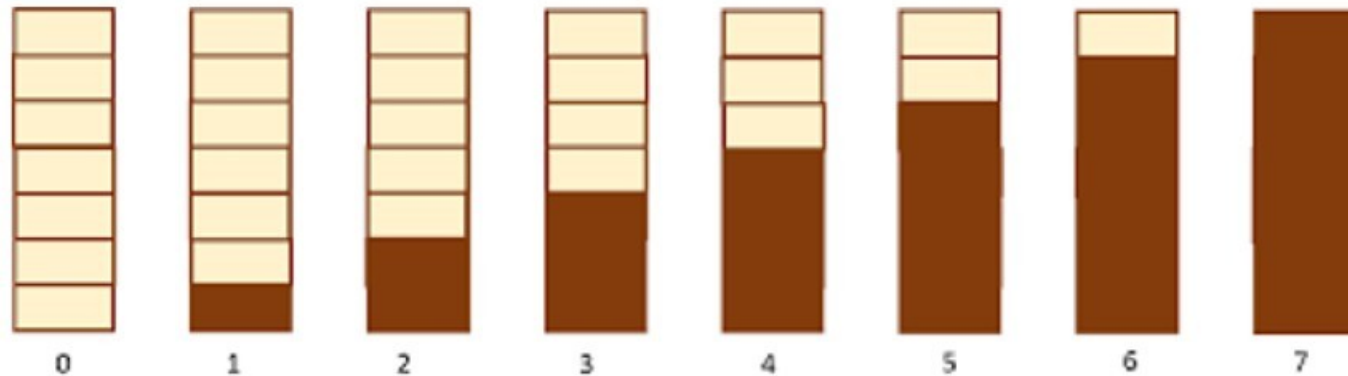
Representing Data

- Numeric: Use a number
 - Blood pressure
 - Temperature
 - Age

Age by Patient		
1	2	3
[23	82	65]

Representing Data

- Thermometer: Convey order, but not spacing or quantity
 - Cancer staging
 - Glasgow Coma Scale
 - RASS Sedation Scale



Cancer Stage by Patient

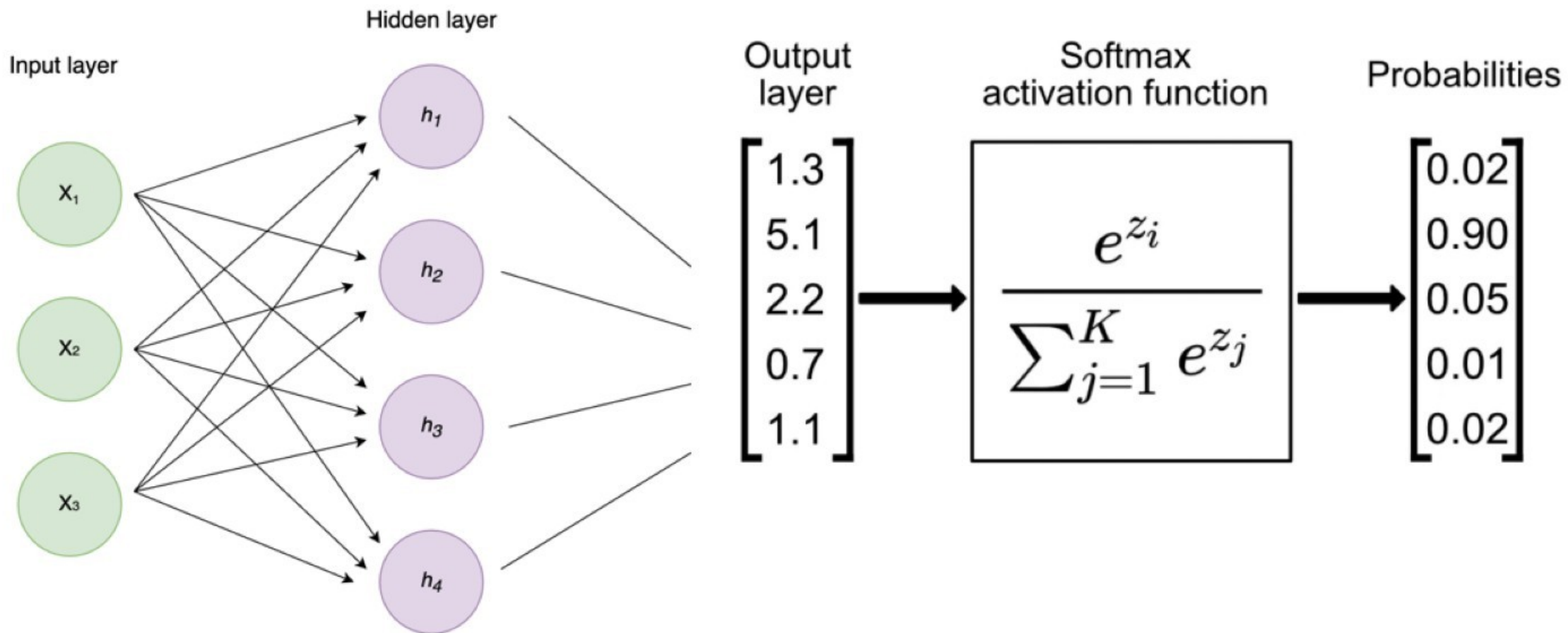
		Patient		
		1	2	3
Stage	I	0	1	1
	II	0	0	1
	III	0	0	1

Representing Data

- One Hot: No meaningful order
 - Hospital Location
 - Insurance Type
 - Drug Allergy
 - Test Results

Location	New York	Los Angeles	Chicago
New York	1	0	0
Los Angeles	0	1	0
Chicago	0	0	1

Softmax Function



Let's Practice

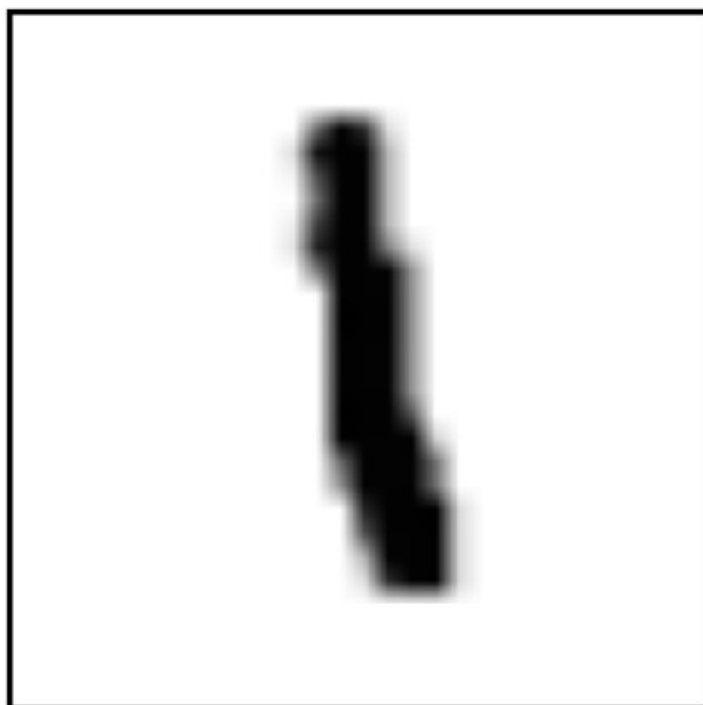
- Amoxicillin 500mg oral TID

Amoxicillin	500mg	Oral	TID
One-hot	Numeric	One-hot	One-hot/ thermometer

- Blood Type

	AB+	AB-	A+	A-	B+	B-	O+	O-
A	1	1	1	1	0	0	0	0
B	1	1	0	0	1	1	0	0
Rh	1	0	1	0	1	0	1	0

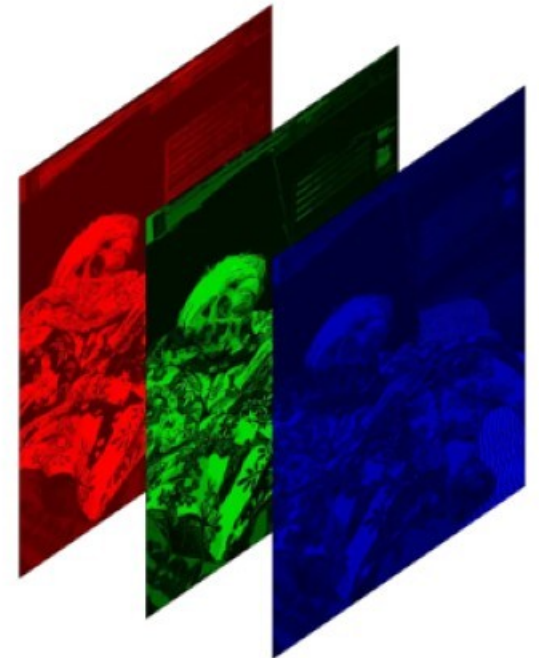
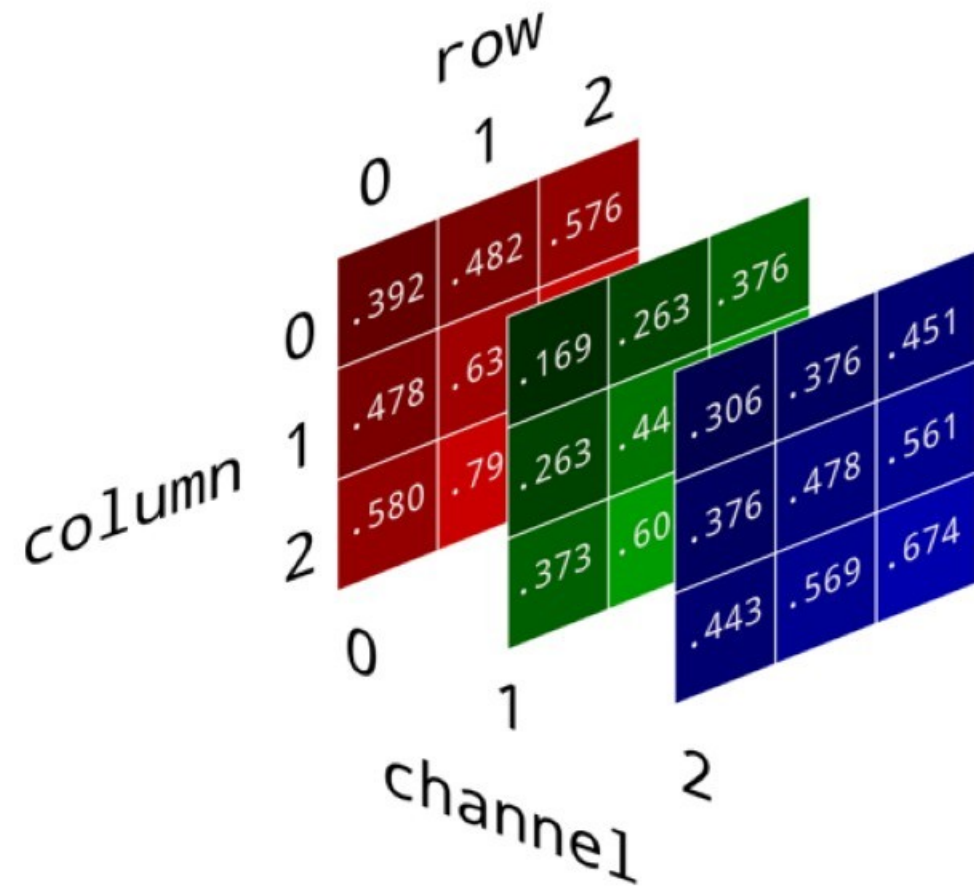
Imaging Data



21

[illegible]

Multi-Channel Images



Bias in Data

- As we attempt to extract patterns from our data we must beware of sources of bias
- Look for patterns:
 - Our data does contain
 - Our data fails to capture
- How we formulate our problem has a massive impact!

Historical Bias

- Patterns in data that reflect inequalities that have occurred as a result of some factor not captured in the model
- 2014 Amazon hiring
- Bias Amplification:
 - Downgrading resumes that mentioned “women’s” (e.g., “women’s chess club captain”)
 - Proxy variables: inner layer predicts man/woman via other variables as output of an internal neuron
- This pattern *is* present but is the result of real world data not reflecting a desired outcome

Sample Bias

- Our sample is not representative of whole population
- Where was the data collected?
 - Community vs academic
 - ICU vs floor
 - Scanners from different vendors

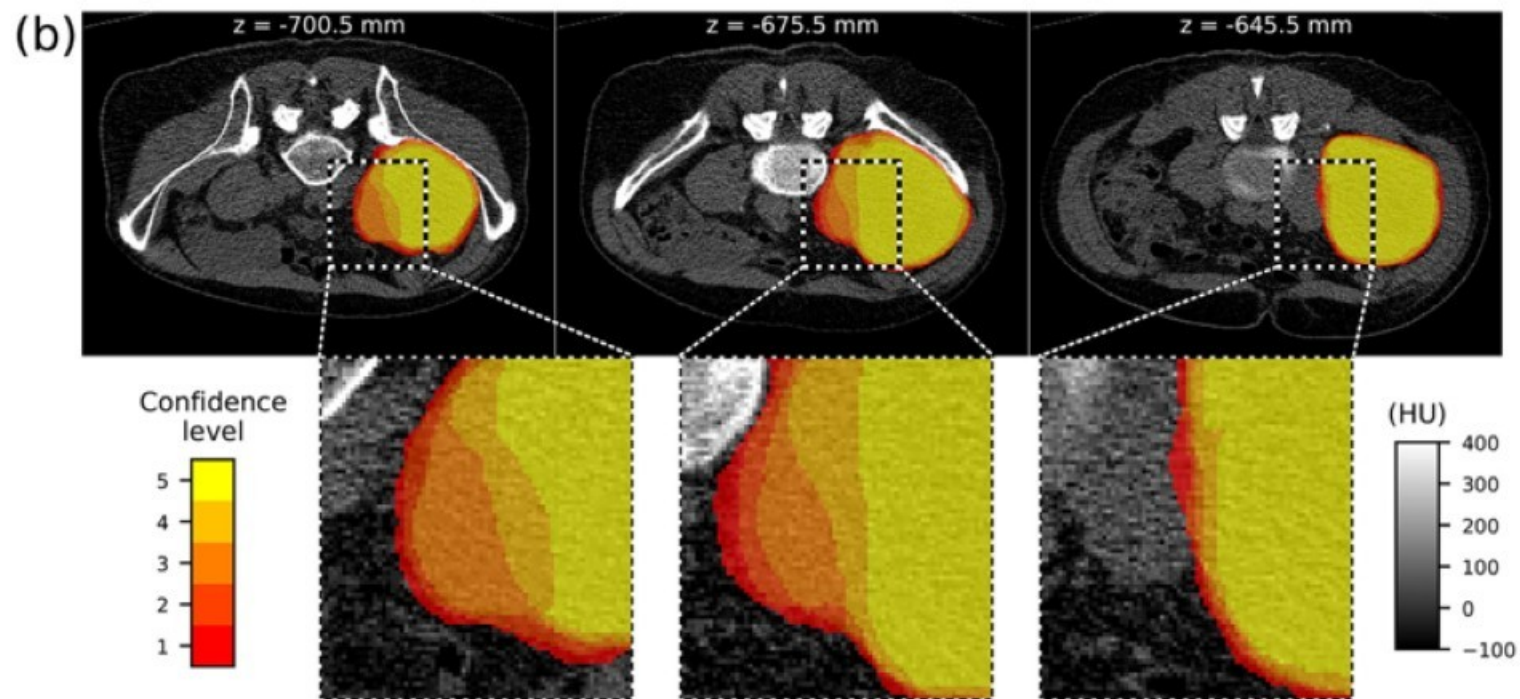
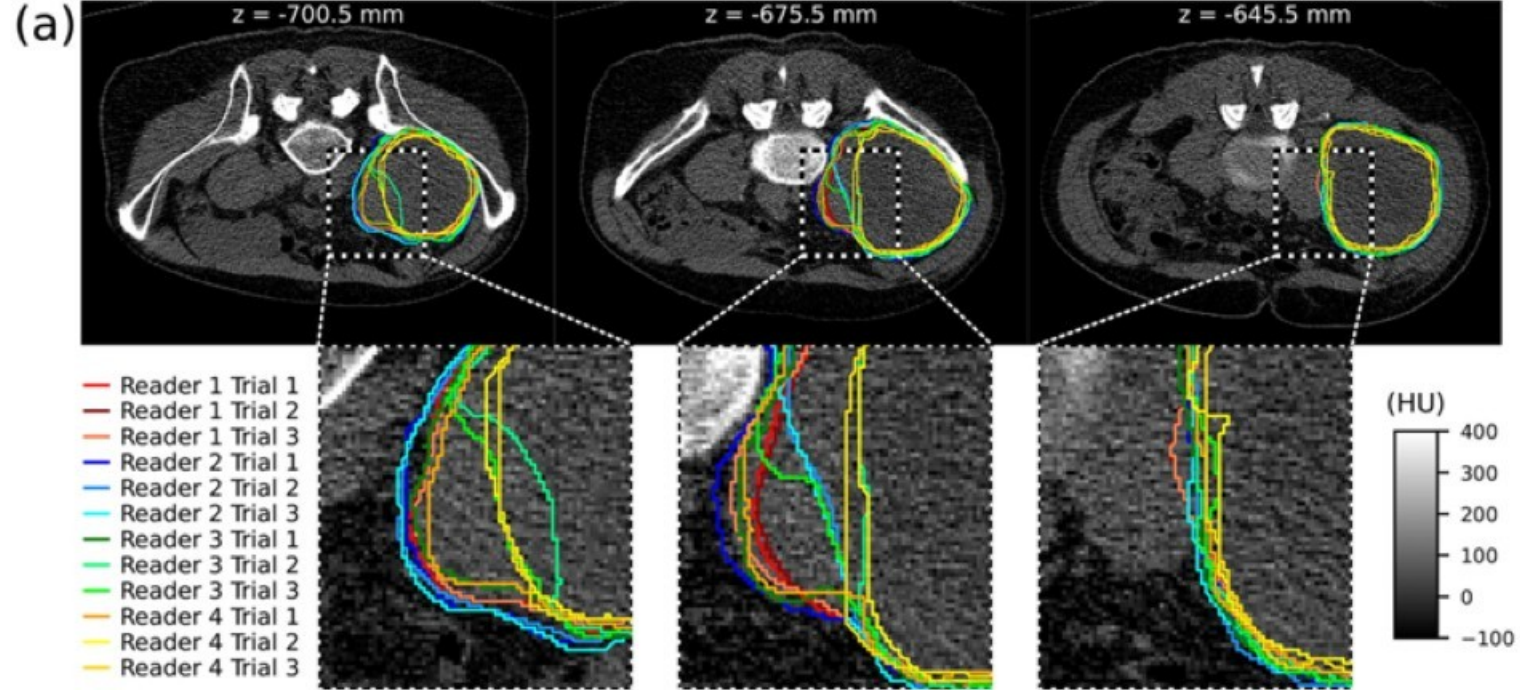
Aggregation Bias

- Arises when clinically meaningful subgroups are collapsed together.
- Imagine if you trained a model on adult and kids x-rays



Label Bias

- Variable labelling by expert annotators
- Ex. Different labelling of tumor radiation treatment zones by different annotators
- Fight by having multiple labelers, have experts reach consensus if disagreement



Data Labelling

- Up to this point, we've discussed supervised learning
- Supervised learning: Where we know the prediction we want to make
- Unsupervised learning: Where the computer spots trends in data according to an algorithm we specify

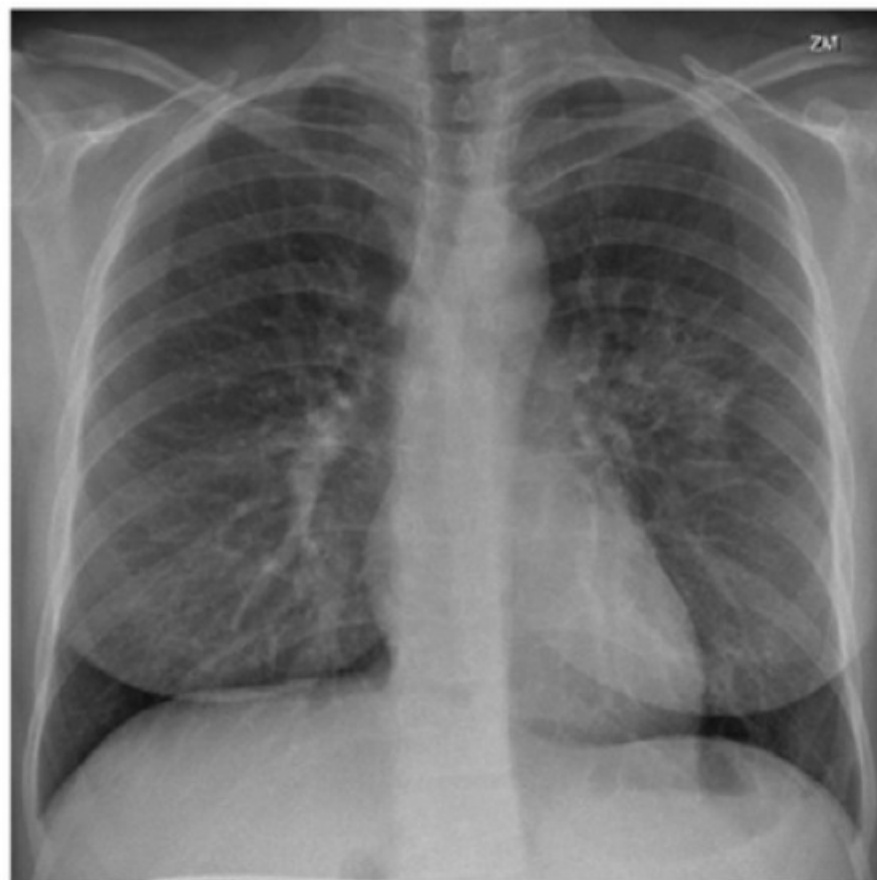
Data Labelling Strategies

- Retrospective study: Label is predicting future value of something that can be recorded in the past
- Human labelers: Crowdsourcing labels (mturk.com/)
 - Label symptoms, medications or diagnoses
- Expert labelers:
 - Medical imaging
 - Cardiac EP
 - EEG

Data Augmentation

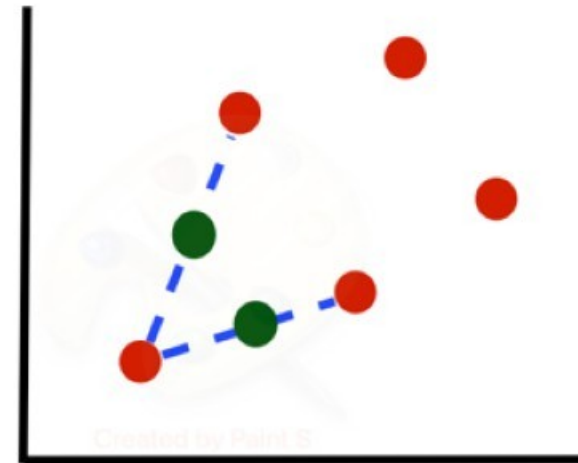
- Introducing noise in your data to prevent signal from being found where there is none
- Changes something about your data that shouldn't matter

Data Augmentations - Radiology



Data Augmentation - SMOTE

- Synthetic Minority Over-sampling TEchnique, or SMOTE
- Create datapoints near those in minority class
- Weights data so that these points have an increased influence on parameter updates during GD/Backprop

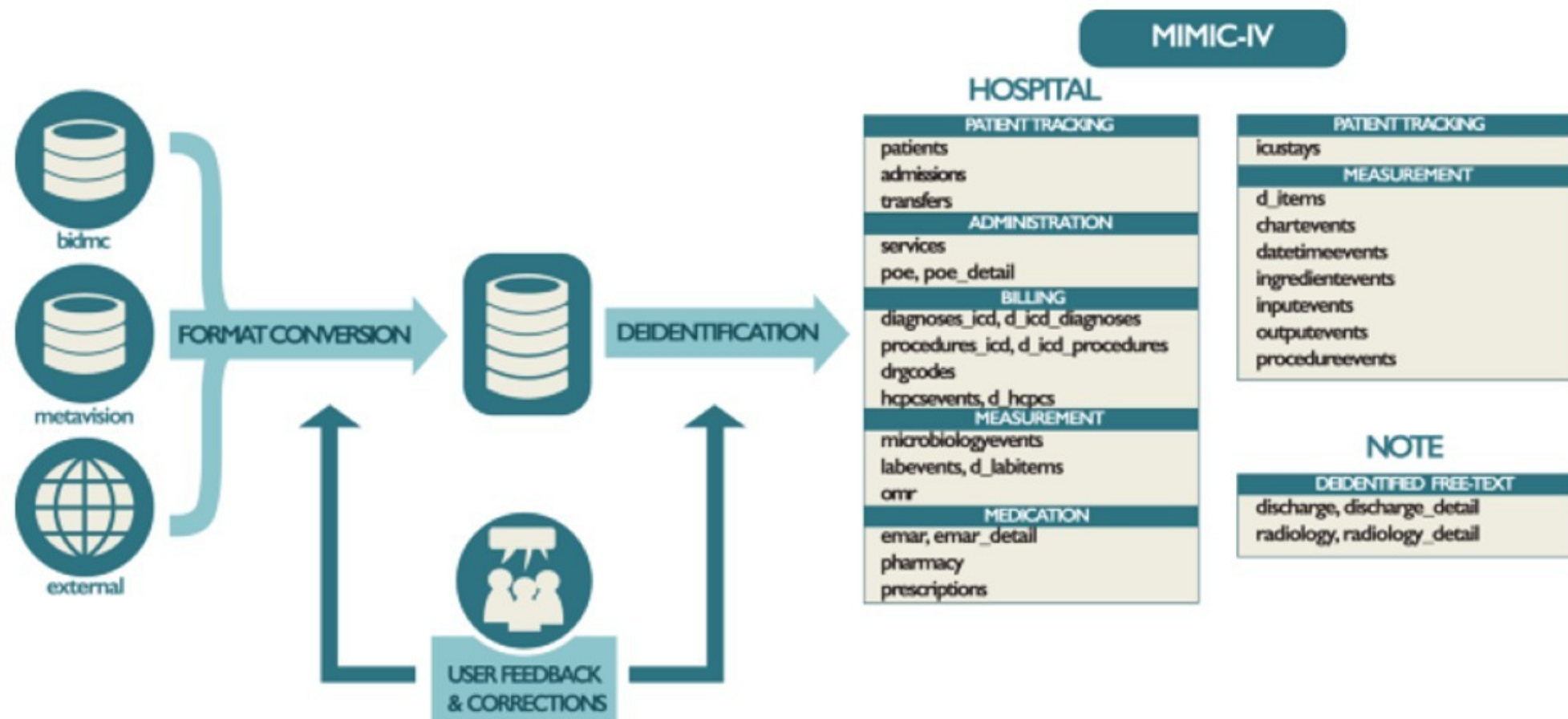


Medical Dataset Challenges

- Data Quality and Completeness
- Heterogeneity of Sources
- Annotation and Labeling
- Scale and Storage
- Privacy and Security
- Bias and Representativeness
- Temporal and Longitudinal Nature
- Rare Events

Medical Datasets - MIMIC

- Medical Information Mart for Intensive Care



NIH Chest X-ray Dataset

- Comprised of 112,120 X-ray images with disease labels from 30,805 unique patients
- Natural Language Processing to text-mine disease classifications from the associated radiological reports
- Not entirely correct (90%)

UK Biobank

- Large-scale biomedical database
- contains genetic, lifestyle, and health information from 500,000+ UK participants, along with medical imaging data.
- Supports research in genomics, personalized medicine, and disease prediction models.



Improving the health of future generations

Kaggle Database/Challenges

RSNA Intracranial Aneurysm Detection

Detect the presence and location of intracranial aneurysms in multimodal imaging data

